SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

ICT 133
Structured Programming

Tutor-Marked Assignment

July 2026 Presentation

TUTOR-MARKED ASSIGNMENT (TMA)

This assignment is worth 24% of the final mark for ICT133 - Structured Programming.

The cut-off date for this assignment is **Sunday, 11 October 2026, 2355hrs**.

Note to Students:

All TMAs must be submitted electronically through Canvas. No partial submission of TMA will be accepted unless otherwise specified.

Assignment Requirements:

- Do **NOT** define classes for this TMA.
- **Unless specified in the question, you CANNOT use packages not covered in this module, e.g., re, collections, numpy, pandas etc.**
- All functions must be documented. Provide sufficient comments to your code and ensure that your program adheres to good programming practices such as not using global variables.
- **Do not use exit() function.**
- Failing to do so can incur a penalty of as much as 50% of the marks allotted.

Submission Details:

- Use the template word document provided - **SUSS_PI_No-FullName_133TMA.docx.** Rename the file with your SUSS PI and full name join with "_133TMA" e.g., "**SUSS_PI-TomTanKinMeng_133TMA.docx**" (without the quotes).
- Include the following particulars on the first page of the word document, on separate lines: **Course Code, SUSS PI No., Your Name, Tutorial Group and Submission Date.**
- Copy and paste the source code of each program you write in **text** format. Submit screenshots for **only** output of your program, where applicable.
- If you submit the source code as a screenshot, your code will not be marked. That is, **screenshot code will be awarded zero mark.**
- Submit your solution in the form of a **single MS Word document. Do NOT submit as a pdf document.** You will be penalised if you fail to submit a word document.
- The word document must be submitted to your respective T group. Besides this, you are required to upload your source code to Vocareum. Do a **print screen** of the Vocareum submission and insert this screenshot into your word document.

Question 1 (15 marks)

This question covers materials in Seminar 1 and 2. Use and express selection structure for this question. Functions, repetition, or collections are not necessary for this question.

EV charging rates at a city charging hub are as follows:

Charger Type	Tier	Rates
DC Fast		$0.68 per kWh
AC Standard	Tier 1: Less than 20 kWh	$0.56 per kWh
	Tier 2: Next 20 – 50 kWh	$0.54 per kWh
	Tier 3: Any energy above 50 kWh	$0.52 per kWh

Assume energy is measured as kWh delivered to the vehicle. Users may charge any decimal amount (e.g. 18.75 kWh).

Write a program that:

i. Prompts for a charger type (AC or DC).
ii. Validates the charger type: if input is not AC or DC, print an error message and end the program
iii. Only if user enters a valid charger type, prompt for energy (kWh) as a non-negative number.
iv. Calculates and displays the amount due:
    - DC Fast: amount = 0.68 x kWh
    - AC Standard (tiered): apply tiered rates to portions of the energy as defined above.
v. Assume user input the correct data type for all input.

Submit and paste screenshots of at least THREE program executions, with different inputs.

Sample 1:
```
Enter charger type (DC/AC): AC
Energy delivered (kWh): 28
Please pay $15.12
```

Sample 2:
```
Enter charger type (DC/AC): DC
Energy delivered (kWh): 15.5
Please pay $10.54
```

Sample 3:
```
Enter charger type (DC/AC): CDC
Wrong charger type, please run the program again
```

(15 marks)

Question 2 (20 marks)

This question covers materials up to Seminar 3. Make use of functions, selection, strings and repetition structures. NO data structures like sets, lists or dictionary should be used for this question. Keep the program modular by defining other functions if necessary.

(a) You are a financial advisor and would like to illustrate the returns of various financial instruments to your client. A financial instrument PlanA will take in a certain deposit (up to a maximum of $30,000) at your client's next birthday and earn interest at a certain rate, compounded until his/her 70th birthday. For example, if interest rate is 6%,

i. At age 55, client puts in $10,000, interest earned for the year will be 6% or $600
ii. At age 56, starting balance will be $10,600, interest earned for the year will be $636 (6% of $10,600)
iii. At age 57, starting balance will be $11,236, interest earned for the year will be $674
iv. Continue until age 70

Write a program to ask the client for inputs of initial deposit amount, age and interest. Check that amount is within 0 to $30,000 (inclusive), age is 1 to 100 (inclusive), and rate is 0 to 10% (inclusive). Print a relevant error message if checks fail, and do not proceed to ask subsequent questions.

Submit and paste screenshots of at least TWO runs with different inputs and print out each scenario in a format similar to below:

```
Sample 1:
Enter initial deposit amount: 10000
Enter age of initial deposit: 55
Enter interest rate in %: 6
Year           Balance  +  Interest  =  End of Year
At age 55       10,000          600          10,600
At age 56       10,600          636          11,236
At age 57       11,236          674          11,910
At age 58       11,910          715          12,625
...

Sample 2:
Enter deposit amount: 50000
Error! Amount must be within 0 to $30,000
```

(10 marks)

(b) Extend part (a) to allow deposits of more than $30,000 but for each year, any balance that is above $30,000 but below $60,000 will earn an interest rate that is 1% lower, and any balance above $60,000 will earn an interest rate that is 2% lower. For example, if interest rate is 6%,

i. At age 55, client puts in $50,000, interest for the year will be 6% of $30,000 plus 5% of $20,000 or $2,800
ii. At age 56, starting balance will be $52,800, interest for the year will be $2,940 ($1,800 + 5% of $22,800)
iii. Continue until age 70

Submit and paste screenshots of at least TWO runs with different inputs and print out each scenario in a format similar to below:

```
Sample 1:
Enter initial deposit amount: 50000
Enter age of initial deposit: 55
Enter interest rate in %: 6
Year           Balance  +  Interest  =  End of Year
At age 55       50,000        2,800          52,800
At age 56       52,800        2,940          55,740
At age 57       55,740        3,087          58,827
...

Sample 2:
Enter initial deposit amount: 220400
Enter age of initial deposit: 55
Enter interest rate in %: 6
Year           Balance  +  Interest  =  End of Year
At age 55      220,400        9,716         230,116
At age 56      230,116       10,105         240,221
At age 57      240,221       10,509         250,729
At age 58      250,729       10,929         261,659
At age 59      261,659       11,366         273,025
At age 60      273,025       11,821         284,846
At age 61      284,846       12,294         297,140
At age 62      297,140       12,786         309,925
At age 63      309,925       13,297         323,222
At age 64      323,222       13,829         337,051
At age 65      337,051       14,382         351,433
At age 66      351,433       14,957         366,391
At age 67      366,391       15,556         381,946
At age 68      381,946       16,178         398,124
At age 69      398,124       16,825         414,949
At age 70      414,949       17,498         432,447
```

(10 marks)

Question 3 (30 marks)

This question covers materials up to seminar 4. Employ structure programming and use of functions to make the program modular.

(a) Extending from Question 2, your client can choose to transition PlanA to the Payout phase between ages 65 to 70 (inclusive). The Payout phase will provide a fixed payout every year on his/her birthday and still earn interest based on the same formulas as in Question 2(b). The fixed yearly payout will continue until there is insufficient balance. For example, if the client chooses to transition at age 65 to start yearly payout of $21,360, with an initial amount of $220,400 deposited at age 55, and an interest rate of 6%,

i. At age 55, client puts in an initial deposit of $220,400 with interest rate of 6%

ii. At age 65, balance would grow to $337,051 when the client receives the first payout of $21,360, after which the remaining $315,691 will earn an interest of $13,528 ($3,300 + 4% of $255,691) for the year
iii. At age 66, after $21,360 payout, the remaining balance will be $307,859 and earn $13,214 ($3,300 + 4% of $247,859) for the year
iv. This will continue until there is insufficient balance to meet the payout amount, when the remaining balance will be returned as the final payout
v. Advise the client at what age he/she will stop receiving payouts

Refactor Question 2's code into functions and write a program to call those functions after prompting your client for new inputs of age to start payout and payout amount. Determine what should be the input and output parameters of the functions and avoid the use of global variables. Try to avoid duplicate codes as much as possible.

Submit and paste screenshots of at least TWO runs with different inputs and print out each scenario in a format similar to below:

Sample 1:

```
Enter age of first payout: 65
Enter yearly payout amount: 21360
Enter initial deposit amount: 220400
Enter age at initial deposit: 55
Enter interest rate in %: 6
Year           Balance  +  Interest  =  End of Year
At age 55      220,400        9,716         230,116
At age 56      230,116       10,105         240,221
At age 57      240,221       10,509         250,729
At age 58      250,729       10,929         261,659
At age 59      261,659       11,366         273,025
At age 60      273,025       11,821         284,846
At age 61      284,846       12,294         297,140
At age 62      297,140       12,786         309,925
At age 63      309,925       13,297         323,222
At age 64      323,222       13,829         337,051

Starting payout with $337,051
Year            Payout  Start of Year    Interest
At age 65       21,360        315,691      13,528
At age 66       21,360        307,859      13,214
At age 67       21,360        299,713      12,889
At age 68       21,360        291,242      12,550
At age 69       21,360        282,432      12,197
At age 70       21,360        273,269      11,831
At age 71       21,360        263,740      11,450
At age 72       21,360        253,829      11,053
At age 73       21,360        243,522      10,641
At age 74       21,360        232,803      10,212
At age 75       21,360        221,655       9,766
At age 76       21,360        210,062       9,302
At age 77       21,360        198,004       8,820
At age 78       21,360        185,464       8,319
At age 79       21,360        172,423       7,797
```

```
At age 80       21,360        158,860       7,254
At age 81       21,360        144,754       6,690
At age 82       21,360        130,084       6,103
At age 83       21,360        114,828       5,493
At age 84       21,360         98,961       4,858
At age 85       21,360         82,459       4,198
At age 86       21,360         65,297       3,512
At age 87       21,360         47,449       2,672
At age 88       21,360         28,762       1,726
At age 89       21,360          9,128         548
Last payout is $9,675 at age 90
```

Sample 2:

```
Enter age of first payout: 70
Enter yearly payout amount: 28560
Enter initial deposit amount: 220400
Enter age at initial deposit: 55
Enter interest rate in %: 6
Year           Balance  +  Interest  =  End of Year
At age 55      220,400        9,716         230,116
At age 56      230,116       10,105         240,221
At age 57      240,221       10,509         250,729
At age 58      250,729       10,929         261,659
At age 59      261,659       11,366         273,025
At age 60      273,025       11,821         284,846
At age 61      284,846       12,294         297,140
At age 62      297,140       12,786         309,925
At age 63      309,925       13,297         323,222
At age 64      323,222       13,829         337,051
At age 65      337,051       14,382         351,433
At age 66      351,433       14,957         366,391
At age 67      366,391       15,556         381,946
At age 68      381,946       16,178         398,124
At age 69      398,124       16,825         414,949

Starting payout with $414,949
Year            Payout  Start of Year    Interest
At age 70       28,560        386,389      16,356
At age 71       28,560        374,185      15,867
At age 72       28,560        361,492      15,360
At age 73       28,560        348,292      14,832
At age 74       28,560        334,563      14,283
At age 75       28,560        320,286      13,711
At age 76       28,560        305,437      13,117
At age 77       28,560        289,995      12,500
At age 78       28,560        273,935      11,857
At age 79       28,560        257,232      11,189
At age 80       28,560        239,861      10,494
At age 81       28,560        221,796       9,772
At age 82       28,560        203,008       9,020
At age 83       28,560        183,468       8,239
At age 84       28,560        163,147       7,426
At age 85       28,560        142,013       6,581
At age 86       28,560        120,033       5,701
```

```
At age 87       28,560         97,174       4,787
At age 88       28,560         73,401       3,836
At age 89       28,560         48,678       2,734
At age 90       28,560         22,851       1,371
Last payout is $24,222 at age 91
```

(15 marks)

(b) Instead of PlanA as specified in part (a), your client can opt for an enhanced PlanB where payouts can continue longer than PlanA.

i. With PlanB, the interest earned during the Payout phase is deposited into a Secondary account separate from the Primary account where payouts are being deducted.
ii. The Primary account will not earn any interest, but the Secondary account will earn the same interest as the combined balance of both Primary and Secondary accounts.
iii. Once the Primary account is depleted, the balance in the Secondary account can be used to buy an insurance policy that will provide payouts until the client passes away.
iv. Advise the client at what age this occurs and the available balance to purchase the insurance policy.

Submit and paste screenshots of at least TWO runs with different inputs and print out each scenario in a format similar to below:

Sample 1:

```
Enter age of first payout: 65
Enter yearly payout amount: 21360
Enter initial deposit amount: 220400
Enter age at initial deposit: 55
Enter interest rate in %: 6
Year           Balance  +  Interest  =  End of Year
At age 55      220,400        9,716         230,116
At age 56      230,116       10,105         240,221
At age 57      240,221       10,509         250,729
At age 58      250,729       10,929         261,659
At age 59      261,659       11,366         273,025
At age 60      273,025       11,821         284,846
At age 61      284,846       12,294         297,140
At age 62      297,140       12,786         309,925
At age 63      309,925       13,297         323,222
At age 64      323,222       13,829         337,051

Starting payout with $337,051
Year            Payout     Primary   Secondary    Interest
At age 65       21,360     315,691           0      13,528
At age 66       21,360     294,331      13,528      13,214
At age 67       21,360     272,971      26,742      12,889
At age 68       21,360     251,611      39,631      12,550
At age 69       21,360     230,251      52,180      12,197
At age 70       21,360     208,891      64,377      11,831
At age 71       21,360     187,531      76,208      11,450
At age 72       21,360     166,171      87,658      11,053
At age 73       21,360     144,811      98,711      10,641
At age 74       21,360     123,451     109,352      10,212
```

```
At age 75       21,360     102,091     119,564       9,766
At age 76       21,360      80,731     129,330       9,302
At age 77       21,360      59,371     138,633       8,820
At age 78       21,360      38,011     147,453       8,319
At age 79       21,360      16,651     155,771       7,797
Exiting Payout phase with $180,220 at age 80
```

Sample 2:

```
Enter age of first payout: 70
Enter yearly payout amount: 28560
Enter initial deposit amount: 220400
Enter age at initial deposit: 55
Enter interest rate in %: 6
Year           Balance  +  Interest  =  End of Year
At age 55      220,400        9,716         230,116
At age 56      230,116       10,105         240,221
At age 57      240,221       10,509         250,729
At age 58      250,729       10,929         261,659
At age 59      261,659       11,366         273,025
At age 60      273,025       11,821         284,846
At age 61      284,846       12,294         297,140
At age 62      297,140       12,786         309,925
At age 63      309,925       13,297         323,222
At age 64      323,222       13,829         337,051
At age 65      337,051       14,382         351,433
At age 66      351,433       14,957         366,391
At age 67      366,391       15,556         381,946
At age 68      381,946       16,178         398,124
At age 69      398,124       16,825         414,949

Starting payout with $414,949
Year            Payout     Primary   Secondary    Interest
At age 70       28,560     386,389           0      16,356
At age 71       28,560     357,829      16,356      15,867
At age 72       28,560     329,269      32,223      15,360
At age 73       28,560     300,709      47,583      14,832
At age 74       28,560     272,149      62,414      14,283
At age 75       28,560     243,589      76,697      13,711
At age 76       28,560     215,029      90,408      13,117
At age 77       28,560     186,469     103,526      12,500
At age 78       28,560     157,909     116,026      11,857
At age 79       28,560     129,349     127,883      11,189
At age 80       28,560     100,789     139,072      10,494
At age 81       28,560      72,229     149,567       9,772
At age 82       28,560      43,669     159,339       9,020
At age 83       28,560      15,109     168,359       8,239
Exiting Payout phase with $191,707 at age 84
```

(15 marks)

Question 4 (35 marks)

This question covers materials up to seminar 4. Employ structure programming and use of functions to make the program modular.

(a) Write a Python function named *openMysteryBox() -> int*: that simulates a game where a player opens mystery boxes to accumulate gold coins into his Treasure Chest. Each box contains either:

- A random number of coins between 1 and 6, or
- A **bomb** (represented by -1), which resets the player's total coins to 0.

The player starts with **0 coins**, and the goal is to reach **exactly 25 coins**. The functional requirements are:

i) The function must be named openMysteryBox()
ii) When a player opens a mystery box, it contains a random number representing a bomb or the number of gold coins in the box:
   a) -1 represents drawing a bomb, the total resets to 0.
   b) Otherwise, the coins (1-6) are added to the Treasure Chest's total only if they do not exceed 25.
   c) If a box contains coins that would cause the total to exceed 25, the coins are discarded, and the player opens another box.
   d) The game ends when the player reaches exactly 25 coins, and the function returns the number of mystery boxes opened.

Note: no user input required.

Submit and paste screenshots of at least TWO sample runs.

Sample 1:

```
Gold Coins: 0
Chest contains: 4 ... Total now: 4
Chest contains: 6 ... Total now: 10
Chest contains: 5 ... Total now: 15
Chest contains: 6 ... Total now: 21
Chest contains: 5 ... Discarded, try again
Chest contains: 4 ... Total now: 25
Treasure Chest Full!
< function returns 6 >
```

Sample 2:

```
Gold Coins: 0
Chest contains: 5 ... Total now: 5
Chest contains: 6 ... Total now: 11
Chest contains: Bomb! Total reset to 0
Chest contains: 5 ... Total now: 5
Chest contains: Bomb! Total reset to 0
Chest contains: 5 ... Total now: 5
Chest contains: 2 ... Total now: 7
Chest contains: Bomb! Total reset to 0
Chest contains: 4 ... Total now: 4
Chest contains: 1 ... Total now: 5
Chest contains: 6 ... Total now: 11
```

```
Chest contains: 4 ... Total now: 15
Chest contains: 5 ... Total now: 20
Chest contains: 5 ... Total now: 25
Treasure Chest Full!
< function returns 14 >
```

(15 marks)

(b) Extend the game in part (a) where **three players** now compete to fill **two Treasure Chests each**. Each chest must reach exactly 25 coins.

You must make use of the openMysteryBox() function from part (a) to simulate the contents of each mystery box.

Game Rules:

i) Players: The game includes "Lara", "Nico", and "Zane".

ii) Initialize a nested list to store each adventurer's progress across two chests:

boxesOpened = [["Lara", 0, 0], ["Nico", 0, 0], ["Zane", 0, 0]]

Each item contains the name and number of boxes opened for each chest. For example, ["Lara",6,8] means Lara opened 6 boxes for the first chest and 8 for the second, totaling 14 boxes.

iii) Mystery Boxes: Each box contains either:
   a) A random number of coins between 1 and 6, or
   b) A bomb (represented by -1), which resets the current chest to 0.

iv) Chest Progression:
   a) Players fill their first chest to exactly 25 before moving to the second. Record the number of boxes opened for this player into the nested list.
   b) If a coin value would cause the current chest to exceed 25, it is discarded.
   c) If a bomb is drawn, the current chest resets to 0.

v) After each player fill up their 2 chests, determine the winner as the player who opened the least number of boxes. Your program must handle the condition that there could be a draw.

Note: no user input required.

Submit and paste screenshots to show TWO sample runs.

Sample 1:

```
…
… output from openMysteryBox() for each player …
…
Game Results:
Lara: Chest 1 = 6, Chest 2 = 8, Total = 14
Nico: Chest 1 = 7, Chest 2 = 9, Total = 16
```

```
Zane: Chest 1 = 5, Chest 2 = 10, Total = 15
Winner: Lara with 14 boxes!

Sample 2:
…
… output from openMysteryBox() for each player …
…
Game Results:
Lara: Chest 1 = 6, Chest 2 = 8, Total = 14
Nico: Chest 1 = 7, Chest 2 = 7, Total = 14
Zane: Chest 1 = 5, Chest 2 = 10, Total = 15
It's a draw between: Lara, Nico with 14 boxes!
```

(20 marks)

---- END OF ASSIGNMENT ----